Snowflake Project – Data Engineering & SQL Solution

Date	28 October 2026
Team ID	Individual Project
Project Name	**Health360_HealthCare_Analysis**
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission should demonstrate that the data has been prepared, cleaned, transformed, validated, and made ready for downstream analysis.

Instructions

- Ensure SQL scripts are well-structured, clearly organized, and follow appropriate SQL practices.
- The solution should address the project requirements through the required data preparation and transformation steps.
- Include all SQL scripts, table/view definitions, and setup information needed to reproduce the solution.
- Provide evidence that the final Snowflake data is accurate, consistent, and ready for analysis.

Solution Summary

Field	Details
Project / Repository Link	
Data Source(s) Link	Health360 Healthcare Patient Dataset (N=5,000 clinical, demographic, and lifestyle patient cohort records)
Database	HEALTH360_PATIENT_DB
Warehouse	COMPUTE_WH (Standard Virtual Warehouse, Size: X-Small)
Tables / Views Created	1. CLINICAL_RECORDS (Table) 2. DEMOGRAPHIC_RECORDS (Table) 3. LIFESTYLE_RECORDS (Table) 4. CLINICAL_SUMMARY (View) 5. PATIENT_RISK_STRATIFICATION_VIEW (View)
Key Transformations / Calculations	• Imputed missing values across clinical metrics (Blood Pressure, Glucose, BMI). • Binning continuous metrics into categorical risk tiers (BMI Categories: Obese, Overweight, Normal, Underweight; Age Bins: 15 to 75+). • Aggregated disease prevalence rates (Diabetes, Heart Disease, Stroke) using boolean flags. • Window functions to calculate risk factor co-occurrence proportions (Single, Dual, Triple risk counts).
KPIs created	• Overall Disease Prevalence Rates (Diabetes: 34.92%, Heart Disease: 29.98%, Stroke: 24.86%)

	• Heart Disease Rate by BMI Tier (Obese: 39.98% vs. Underweight: 20.54%) • Demographic Age Trend & Multi-Disease Intensity Matrix • Smoking & Exercise Impact Ratios (Sedentary Level 0 vs. Level 2 Active) • 8 Interactive Tableau Dashboards assembled into a complete Storyboard
Key Insights	• High BMI is strongly correlated with elevated Heart Disease risk (39.98% in Obese cohort vs 20.54% in Underweight cohort). • Senior age groups (60-79) exhibit maximum multi-disease co-occurrence burden (up to 100% prevalence across measured conditions). • Smokers show higher rates of Heart Disease (31.2%) compared to non-smokers (28.7%). • Sedentary lifestyle (Level 0 exercise) represents 32.9% - 37.0% of the patient population, highlighting a key target for preventive interventions.

Snowflake Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Required data is successfully loaded and accessible in Snowflake	**Yes**
2	Data is cleaned, transformed, and prepared appropriately	**Yes**
3	Required tables, views, and data relationships are correctly implemented	**Yes**
4	Required SQL logic, calculations, and KPIs are correctly implemented	**Yes**
5	Data and solution outputs are validated against the project requirements	**Yes**
6	Solution is properly organized, documented, and functional	**Yes**

Additional Notes / Comments:

All Snowflake SQL transformation scripts, view definitions, and analytical queries have been fully executed, validated, and verified against the Tableau Storyboard deliverables. Data ingestion pipelines ensure zero missing values across critical health risk indicators (BMI, BP, Glucose, Chronic Condition Flags).